# Md. Juwel Ahmed Sarker

212 Graduate Education Building, University of Arkansas, Fayetteville, AR 72701 | +1 (479) 332-9288
sarker@uark.edu | ORCID | Google Scholar | website: https://mdjuwelsarker.github.io/

## RESEARCH INTERESTS

Higher Education Policy; Student Well-Being and Graduate Employability; School Finance and Governance; Economics of Education; Teacher Labor Markets; Applications of Machine Learning to Education Policy.

## EDUCATION

Expected 2027 **Ph.D. in Education Policy, University of Arkansas**
Dissertation: *"Making School Dollars Meaningful: Consolidation, Efficiency, and the Economics of School Finance"*
Committee: Josh B. McGee (Chair), Harry Patrinos, Robert Maranto

2018 M.Econ, Development Economics, Dhaka School of Economics

2012 M.S.S., Development Studies, University of Dhaka

2011 B.S.S. (Honors), Development Studies, University of Dhaka

## JOB MARKET PAPER

*"When Small Districts Disappear: Fiscal Consequences of Consolidation in Arkansas,"* with Josh B. McGee.
Presented at AEFP 51st Annual Conference, Chicago, 2026; at SREE 2026 and the ABFM 38th Annual Research Conference, Washington, DC, September 2026.

Featured in **The Hechinger Report:** Jill Barshay, "Merging school districts was supposed to save taxpayers money. It didn't, a 20-year study in Arkansas found," October 5, 2026.

## ACADEMIC APPOINTMENTS AND PROFESSIONAL EXPERIENCE

### Academic Appointments

2023–Present Graduate Assistant, University of Arkansas

2019–Present Assistant Professor (on leave), Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh

2021–23 Chairman, Department of Development Studies, Hajee Mohammad Danesh Science and Technology University

2022–23 Assistant Proctor, Hajee Mohammad Danesh Science and Technology University

2021–22 Assistant Hall Superintendent, Hajee Mohammad Danesh Science and Technology University

### Professional Experience

2015–2019 Program Officer, Skills for Employment Investment Program (SEIP), Palli Karma-Sahayak Foundation

2013–2015 Research Associate, Microcredit Regulatory Authority

2015 Junior Consultant, National Land Zoning Project, Ministry of Land, Bangladesh

**Research Assistantships**

2012 Field Research Assistant, Bangladesh Enterprise Institute (BEI), "Local Government Audit and Accountability Systems" Project

2012 Intern, UNDP, Dhaka, Bangladesh, in the Poverty Environment and Climate Mainstreaming Project (PECM)

2011 Field Research Assistant, Department of Development Studies, University of Dhaka, in the "Dynamics of Urban Poverty" project (under Prof. M. Abu Eusuf)

2009–2011 Research Assistant, Institute for Research and Development (IRD), Dhaka

## PUBLICATIONS

† denotes student mentee.

**Peer-Reviewed Journal Articles**

2026 **Sarker, M. J. A.**, Maranto, R., Haque, A.†, & Sagar, M. S. U.† (2026). Social media impacts on Bangladeshi college students. *Discover Psychology*.

2025 **Sarker, M. J. A.**, Hasan, M., Kabir, A.†, & Haque, A.† (2025). Leveraging artificial intelligence to assess the impact of COVID-19 on the teacher-student relationship in higher education. *PLOS ONE*, 20(3), e0317567.

2022 Hossain, M. T., Islam, M. A., Jahan, N., Nahar, M. T., **Sarker, M. J. A.**, Rahman, M. M., ... & Islam, M. N. (2022). Mental health status of teachers during the second wave of the COVID-19 pandemic: a web-based study in Bangladesh. *Frontiers in Psychiatry*, 13, 938230.

Hossain, M. T., Ahammed, B., Jahan, N., Islam, M. A., Rahman, M. M., Khan, B., **Sarker, M. J. A.,** ... & Islam, M. N. (2022). Prevalence and predictors of pornography exposure during the third wave of the COVID-19 pandemic: A web-based cross-sectional study on students in Bangladesh. *Frontiers in Public Health*, 10, 1046813.

**Book Chapters**

2020 **Sarker, M. J. A.** (2020). The readymade garment sector: Vulnerabilities and resilience. In R. A. M. Titumir, N. Georgeou, & A. Chowdhury (Eds.), *COVID-19 and Bangladesh: Response, Rights, and Reliance* (pp. 75–81). University Publication Limited.

**Under Review**

**Sarker, M. J. A.**, Hasan, M., Kabir, A.†, Haque, A.†, & Hamja, A. (under review). A data-driven analysis of the education-employment linkage with explainable artificial intelligence. *Telematics and Informatics Reports*.

## MANUSCRIPTS IN PREPARATION

Sarker, M. J. A., & McGee, J. B. (n.d.). *"Merging for Efficiency? Causal Evidence from Arkansas School District Consolidation."*

Presented under an earlier working title ("Assessing Relative Educational Efficiency of Arkansas School Districts Using a Modified Simar-Wilson Two-Stage DEA") at AEFP 50th Annual Conference, Washington, DC, 2025, and (as "Measuring Efficiency of School Districts in Arkansas: A Data-Driven Approach") at SREE 2025.

Sarker, M. J. A., McGee, J. B., & Maranto, R. (n.d.). *"The Leadership Multiplier: How Leadership Shapes the Meaning of School Dollars."*

(Manuscript drafted).

Sarker, M. J. A., Sagar, M. S. U.†, Tabassum, A., & Islam, T. (n.d.). *"Cannot Read a Word: Predictors of Non-Reading and the Literacy Gap in Rural Bangladesh."*
(Manuscript drafted).

McGee, J. B., Sarker, M. J. A., Zamarro, G., & Camp, A. (n.d.). *"If at First You Don't Succeed: The Impact of Failing the First Attempt on a Teacher Licensure Exam on Licensure, Employment, and Value-Added."*
(Manuscript drafted and Presented under evolving working titles at AEFP 49th Annual Conference, Baltimore, MD, March; SREE 2024; and the APPAM Fall Research Conference, National Harbor, MD, November).

Sarker, M. J. A., Sagar, M. S. U.†, Tofayel, A. H., Nahid, F., & Jahan, S. (n.d.). *"Stuck on Mobile: Device Access and the Gender Digital Skills Gap."*
(Manuscript drafted).

Wolf, P., Shakil, D., Patrinos, H., Daniels, D., Wilson, T., Helal, A. M., & Sarker, M. J. A. (n.d.). *"The Impact of Private School Choice on Student Attainment around the Globe: A Meta-Analysis & Returns to Education Study."*
Data analysis underway.

Helal, A. M., Wai, J., Sarker, M. J. A., & Rieske, A. (n.d.). *"The Returns to Gifted Education: A Meta-Analysis of Academic and Long-Term Outcomes Across Research Designs."(*Initial draft complete).

## CONFERENCE PRESENTATIONS

2026

Sarker, M. J. A., & McGee, J. B. (2026, September). *When small districts disappear: The fiscal consequences of consolidation in Arkansas* [Paper presentation]. SREE 2026 Conference.

Sarker, M. J. A., & McGee, J. B. (2026, September). *When small districts disappear: The fiscal consequences of consolidation in Arkansas* [Paper presentation]. ABFM 38th Annual Research Conference, Washington, DC.

Sarker, M. J. A., & McGee, J. B. (2026). *When small districts disappear: Fiscal consequences of consolidation in Arkansas* [Paper presentation]. AEFP 51st Annual Conference, Chicago, IL.

2025

Sarker, M. J. A., & McGee, J. B. (2025, September). *Measuring efficiency of school districts in Arkansas: A data-driven approach* [Paper presentation]. SREE 2025 Conference.

Sarker, M. J. A., & McGee, J. B. (2025). *Assessing relative educational efficiency of Arkansas school districts using a modified Simar-Wilson two-stage Data Envelopment Analysis (DEA)* [Paper presentation]. AEFP 50th Annual Conference, Washington, DC.

Sarker, M. J. A., & Maranto, R. (2025). *Will social media contagion harm teens in non-Western societies? Experience of Bangladesh* [Poster presentation]. AEFP 50th Annual Conference, Washington, DC.

2024

McGee, J. B., Sarker, M. J. A., Zamarro, G., & Camp, A. (2024, November). *If at first you don't succeed: The impact of failing the first attempt on a teacher licensure exam on licensure, employment, and value-added* [Paper presentation]. APPAM Fall Research Conference, National Harbor, MD.

McGee, J. B., Sarker, M. J. A., Zamarro, G., & Camp, A. (2024, September). *Assessing the impact of teacher licensure exam scores on licensure, employment as teacher, and teaching effectiveness* [Paper presentation]. SREE 2024 Conference.

McGee, J. B., Sarker, M. J. A., Zamarro, G., & Camp, A. (2024, March). *Assessing the relationship between teacher licensure exam scores and classroom effectiveness* [Paper presentation]. AEFP 49th Annual Conference, Baltimore, MD.

## FELLOWSHIPS AND AWARDS

| | |
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| 2026–2027 | School Finance Fellow, EdFund |
| 2023–Present | Doctoral Academy Fellow, University of Arkansas |
| 2022 | Best Research Paper Presenter (Social Science Faculties), Institute for Research and Training (IRT), Hajee Mohammad Danesh Science and Technology University |
| 2012 | Governor Scholarship, Bangladesh Bank |
| 2011 | Development Studies Scholarship, University of Dhaka |
| 2009–10 | Genius Scholarship, Centre for Zakat Management (CZM), Dhaka |

## GRANTS AND FUNDED PROJECTS

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| 2025 | Research Travel Grant, Graduate-Professional Student Congress (GPSC), University of Arkansas. |
| 2023 | Principal Investigator, *"Digital Bangladesh and Youth Employment: How Can Digital Competence Enhance Employability of Tertiary Graduates in Bangladesh?"* University Grants Commission, Bangladesh. |
| 2022–23 | Principal Investigator, *"Connecting Life: Predicting and Preventing Suicide among University Students through Machine Learning Methods,"* Institute of Research and Training. |
| | Principal Investigator, *"What Causes Unemployment among Educated Youths in Bangladesh? Demystifying the Psycho-Social and Socio-Economic Narratives,"* Institute of Research and Training. |
| 2021–22 | Evaluation Expert, *Impact Evaluation of e-Learning Pilot Project,* UNDP Bangladesh. |
| 2021 | Principal Investigator, *"Linking Education to Employment in Bangladesh: Where to Focus?"* Social Science Research Council, Ministry of Planning, Bangladesh. |
| 2020–21 | Principal Investigator, *"Operational Efficiency of and Financial Inclusion in the Microfinance Sector in Bangladesh: A Panel Analysis,"* Institute of Research and Training, HSTU. |
| 2019–20 | Principal Investigator, *"Perception and Practice of Freedom and Independence among Working Women in Bangladesh,"* Institute of Research and Training, HSTU. |

## TEACHING EXPERIENCE

**Department of Education Reform, University of Arkansas**

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| 2024 | Teaching Assistant, EDRE 61003: Quantitative Analytical Techniques for Education Policy, Fall 2024 (led Stata labs, graded quantitative assignments and projects, and provided methodological mentoring) |

**Hajee Mohammad Danesh Science and Technology University, Dinajpur, Bangladesh**

2023 DEV 125: Technological Revolutions and Development, Spring 2023

DEV 126: Computing and Data Analysis Skills for Development (Lecture and Lab), Spring 2023

DEV 215: Environment, Development and Sustainability, Spring 2023

DEV 225: NGOs and Development Management, Spring 2023

DEV 325: Education, Health, Labour, and Human Development, Spring 2023

ECN 251: Mathematical Economics, Spring 2023

SOC 413: Social Policy and Planning, Spring 2023

2022 DEV 125: Technological Revolutions and Development, Spring 2022

DEV 126: Computing and Data Analysis Skills for Development (Lecture and Lab), Spring 2022

DEV 215: Environment, Development and Sustainability, Spring 2022

DEV 225: NGOs and Development Management, Spring 2022

DEV 313: Poverty and Inequality: Concept, Measurement and Policy, Fall 2022

DEV 317: Banking, Finance and Insurance, Fall 2022

ECN 301: Development Economics, Fall 2022

2021 ECN 125: Macroeconomics, Spring 2021

2020 ECN 129: Economics, Fall 2020

ECN 125: Macroeconomics, Spring 2020

2019 ECN 125: Macroeconomics, Spring 2019

ECN 307: History of Economic Thought, Spring 2019

Student feedback: instructor-run surveys, ECN 301 Development Economics (n = 41), mean rating 9.1 out of 10.

**Pedagogical Training**

2022 Objective-Based Curriculum Development and Classroom Management, Hajee Mohammad Danesh Science and Technology University.

2018 Overseas Pedagogy Training for Bangladeshi TVET Trainers: Student-Centered Lesson Plan Development and Classroom Management, Institute of Technical Education – Education Services, Singapore (June 17–30).

## MENTORING

**Research mentoring:** Mentored more than 15 senior undergraduate students through research projects, including funded studies of graduate employment and student well-being.

**Student co-authored publications:** Two peer-reviewed articles co-authored with mentees (PLOS ONE, 2025; Discover Psychology, 2026); a third is under review and further manuscripts with mentees are drafted.

**Mentee outcomes:** At least ten mentees have advanced to further study or academic positions: a PhD program (United States), MS programs (Germany, Italy), MA programs (India, Thailand), and one faculty appointment at Hajee Mohammad Danesh Science and Technology University.

## SERVICE

### Manuscript Reviewer

Humanities & Social Sciences Communications (Nature Portfolio), 2025 • Gifted Child Quarterly, 2025–2026 • PLOS ONE, 2022 • International Journal of School Choice, 2026

### Professional Memberships

Association for Education Finance and Policy (AEFP) • Society for Research on Educational Effectiveness (SREE) • Association for Budgeting and Financial Management (ABFM) • Lifetime Registered Graduate, University of Dhaka. Past: APPAM, AERA.

### University Service

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|---|---|
| 2025–Present | Departmental Student Organization Representative, Department of Education Reform (EDRE), University of Arkansas |
| 2025 | Department Representative, Department of Education Reform, Graduate-Professional Student Congress (GPSC), University of Arkansas |

### Departmental Leadership

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| 2021–23 | DevTalk — initiated an academia–industry collaboration through student fieldwork placements, internships, and a workshop series with visiting professionals, as Chairman, Department of Development Studies, Hajee Mohammad Danesh Science and Technology University |
| 2020 | Led undergraduate curriculum development, Department of Development Studies, Hajee Mohammad Danesh Science and Technology University |

### Leadership Initiatives

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| 2020–21 | Founder, Bangladesh Open Learning Society (BOLS) — an online learning community of over 10,000 members who completed more than 70,000 hours of coursework and earned 4,500 course certificates (Coursera, edX) during the COVID-19 pandemic |

### Community Service

Backpacks of Hope, distribution of educational materials to disadvantaged children (2023) • General Secretary, Kabi Jasim Uddin Hall Debating Club, University of Dhaka • Team Lead, University of Dhaka Debate Team, National and International Debate Championships, incl. Quarterfinalist, 2nd OIC Intervarsity Debate Tournament (2012) • Blood Donor, Badhan, University of Dhaka

## SKILLS

**Causal Inference & Econometric Methods:** Randomized Controlled Trials (RCT), Regression Discontinuity Design (RDD), Difference-in-Differences (DID), Instrumental Variables (IV)

**Quantitative Methods:** Hierarchical Linear Modeling (HLM), Structural Equation Modeling (SEM), Machine Learning / Explainable AI, Data Envelopment Analysis (DEA), Stochastic Frontier Analysis (SFA)

**Survey Design & Field Data Collection:** questionnaire design, enumerator training and supervision, baseline/endline survey implementation

**Qualitative Research Methods:** life history method, in-depth interviews (IDI), focus group discussions (FGD), key informant interviews (KII)

**Statistical/Programming:** Stata, R, Python

**Other:** MS Project